

SHIWAM KUMAR

SOC Analyst | VAPT & API Security | AI Security Engineer (Aspiring)

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PROFESSIONAL SUMMARY

SOC Analyst with 1+ year at Skillmine Technology, deployed onsite to the National Informatics Centre (NIC), Govt. of India. Triage 150–200+ daily alerts on ArcSight SIEM and run L1 incident response per NIST 800-61. EC-Council CSA certified with hands-on VAPT/API security experience — found Critical SQLi, BOLA, and privilege-escalation flaws across web, API, and AI-application environments. Targeting SOC L2, VAPT, and AI Security Engineer roles.

TECHNICAL SKILLS

SIEM & Monitoring: ArcSight (primary), Splunk, Microsoft Sentinel, ELK Stack

Incident Response & Threat Intel: NIST 800-61 IR Lifecycle, Alert Triage, Log Correlation, MITRE ATT&CK, Cyber Kill Chain, IOC Enrichment (VirusTotal, AbuseIPDB, URLScan)

Detection & Analysis: Sigma Rules, Suricata, Wireshark, Phishing Triage, Static Malware Analysis (PE Headers, Strings, Hash Lookups), Sandbox Detonation (Any.run)

VAPT & API Security: OWASP Top 10, OWASP API Security Top 10, BOLA/IDOR Testing, JWT & Auth Testing (Algorithm Confusion), Burp Suite, OWASP ZAP, Nmap, Nessus, SQLMap, Metasploit (basics), Hydra

AI/LLM Security: OWASP Top 10 for LLM Applications, Prompt Injection Testing, Secure AI API Design, LLM Guardrails, Rate Limiting & Audit Logging

Scripting & Platforms: Python (FastAPI, log parsing, automation), Bash, PowerShell (basics), Linux (Kali, Ubuntu), Windows Server, Active Directory basics, pfSense

PROFESSIONAL EXPERIENCE

SOC Analyst — Level 1, National Informatics Centre (NIC), Govt. of India

May 2025 – Present

Employed via Skillmine Technology · ArcSight SIEM · EDR · MITRE ATT&CK · Incident Response

- Triaged 150–200+ daily alerts on ArcSight SIEM; tuned 12+ correlation rules across endpoint and firewall use cases, reducing false-positive volume by ~25% in Q3 2025.
- Executed L1 incident response per NIST 800-61 (identification, containment, eradication, escalation); reduced mean time to escalate (MTTE) for P1 incidents from ~45 to ~18 minutes.
- Investigated 30+ phishing cases monthly — validated IOCs (URLs, file hashes, sender IPs) against VirusTotal, AbuseIPDB, and URLScan; coordinated URL/IP blocks with the network team.
- Performed static malware analysis (PE headers, strings, hash lookups) and dynamic detonation in sandbox VMs; mapped findings to MITRE ATT&CK techniques (T1059, T1071) for L2 escalation context.
- Authored a reusable phishing triage playbook adopted into live NIC SOC operations; maintained 100% SLA compliance on L1 alert resolution.

Cybersecurity Intern, Tata Steel Utilities & Infrastructure Services Ltd.

Aug 2023 – Oct 2023

Vulnerability Assessment · Secure Development · OWASP · Nessus

- Assessed internal web applications using OWASP Top 10 methodology; identified 5+ medium-severity issues (weak input validation, missing security headers, verbose errors) and remediated SQLi/XSS via parameterized queries, sanitization, and CSP. Ran Nmap/Nessus scans across internal assets, flagging open ports and misconfigurations.

PROJECTS

Web Application VAPT — E-commerce Test Environment · Burp Suite, OWASP ZAP, SQLMap, Nmap

- Conducted end-to-end web app pentest per OWASP Testing Guide v4. Identified Critical SQLi in login endpoint enabling authentication bypass (CVSS 9.1) and reflected XSS in search parameter exposing session tokens; documented PoCs and recommended parameterized queries, CSP, and output encoding fixes.

API Security Assessment — Multi-Tenant IP Reputation Platform (Black-Box) · cURL, JWT Analysis, OpenAPI Enumeration

- Performed an authorized black-box pentest of a multi-tenant FastAPI/REST API; enumerated 20+ endpoints via exposed OpenAPI schema and found a Critical BOLA flaw allowing cross-tenant access to other orgs' API keys/secrets (CVSS 9.1).
- Confirmed mass-assignment privilege escalation and unauthenticated tenant enumeration; delivered a professional VAPT report with CVSS scoring, PoCs, and remediation aligned to the OWASP API Security Top 10.

Secure AI API & Automated LLM Vulnerability Scanner · Python, FastAPI, JWT, PyJWT

- Built a secure FastAPI application with JWT authentication, RBAC, prompt-injection/SQli input guardrails, rate limiting, and audit logging aligned to the OWASP Top 10 for LLM Applications; then built “LLMScan,” a Python scanner that automatically tests LLM-backed APIs across 9 attack categories and validates detection accuracy against hardened vs. intentionally vulnerable targets.

FamPay CTF — JWT Authentication Bypass (Public Security Challenge) · ctf.fampay.co, JWT Analysis, Burp/cURL

- Exploited a JWT ‘alg:none’ algorithm-confusion flaw on FamPay’s public CTF to bypass signature verification and forge an admin-role token, escalating from user to full admin access; captured the flag and reported the fix (pinned server-side algorithm allow-list).

CERTIFICATIONS & EDUCATION

Certifications: EC-Council CSA, 2026 | Cisco CyberOps Associate, 2026 | Cisco Cybersecurity Essentials, 2025 | CompTIA Security+ (In progress)

Education: B.Tech, Computer Science & Engineering, RVS College of Engineering & Technology (2020–2024) | CGPA: 7.98 / 10.0